Performance Testing

Date	01 October 2026
Team ID	SWTID-2026-6912
Project Name	FitBuddy - AI Fitness Plan Generator using Gemini Models
Maximum Marks	5 Marks

Performance Testing

Performance testing for FitBuddy is intended to evaluate the behavior of the FastAPI web application, AI-powered workout generation, nutrition-tip generation, feedback-based plan updates, database operations, and local deployment under expected usage conditions. The supplied project document describes local testing and verification, but does not provide measured response-time, throughput, CPU, memory, or error-rate results; therefore actual values are marked as pending rather than invented.

Step 1: Testing Overview

Field	Details
Testing Tool Used	FastAPI interactive API documentation (/docs) and browser-based local testing; no dedicated load-testing tool is specified in the project document.
Type of Testing	Functional performance verification / local deployment testing. Load and stress measurements are planned but not documented as completed.
Target Module / API	/generate-workout, /submit-feedback, /view-all-users, and the home page /.
Test Environment	Local development server using FastAPI and Uvicorn at http://127.0.0.1:8000.
Test Date	01 October 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Generate a personalized 7-day workout plan and nutrition tip through /generate-workout using valid user details.	1	60	Request completes successfully and returns the generated plan and nutrition tip.
2	Submit feedback through /submit-feedback to update an existing workout plan.	1	60	Original plan is retrieved, updated through Gemini, stored, and displayed.
3	Open /view-all-users and retrieve stored users and original/updated plans.	5	60	Admin view loads database records correctly without request failure.

4	Concurrent workout-generation requests to assess behavior under increased usage.	10	120	Application remains responsive; errors and resource usage remain within acceptable limits.

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Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	Pending measurement	Pending	No measured average response time is provided in the project document.
2	Response Time (Max)	< 5 seconds	Pending measurement	Pending	Requires recorded maximum response time during a load test.
3	Throughput (Req/sec)	To be measured	Pending measurement	Pending	Measure requests processed per second under the selected workload.
4	Error Rate	< 1%	Pending measurement	Pending	Verify failed API requests and application errors during testing.
5	CPU Utilization	< 80%	Pending measurement	Pending	Monitor local server CPU utilization during concurrent requests.
6	Memory Utilization	< 80%	Pending measurement	Pending	Monitor memory utilization during AI generation and database operations.

Step 4: Observations & Analysis

Key Findings:
The supplied project document confirms that FitBuddy can be run locally with FastAPI/Uvicorn, exposes interactive API documentation, and supports workout generation, nutrition tips, feedback-based plan updates, and database-backed user/plan views. Quantitative performance results are not included in the supplied document.

Bottlenecks Identified:
Potential performance-sensitive areas are Gemini API calls for workout generation and feedback updates, database operations, and concurrent requests. These are areas to monitor during actual load testing; no measured bottleneck is documented.

Optimization Steps Taken:
The project uses separate Gemini Pro and Flash functions for different tasks, FastAPI routing, SQLite/SQLAlchemy persistence, and a modular template structure. These design choices are documented, but no measured performance improvement is reported.

Step 5: Screenshots / Evidence

Evidence 1

FastAPI interactive API documentation (/docs) screenshot to be inserted here.

Evidence 2

Local FitBuddy application / generated workout page screenshot to be inserted here.

Team Work Allocation

Team Member	Performance Testing Responsibility
Pavithra M	Test /generate-workout response behavior and workout-generation flow.
Manisha S	Test nutrition-tip generation and response behavior of Gemini Flash integration.
Monica G	Test /submit-feedback and performance of feedback-based plan updates.
Monisha Anandan	Test database operations, /view-all-users, and backend API behavior.
Mridula M	Test frontend responsiveness, local deployment, evidence screenshots, and final performance report.